ENTERPRISE ENGINEERING GUIDE

Apache Guacamole Enterprise VDI Gateway

Production Deployment, Active Directory & LDAP Authentication, TOTP MFA, SSL/WebSocket Reverse Proxy, and Session Delivery for Proxmox VDI

Document Purpose:	Production VDI Gateway Implementation Runbook
Hypervisor Platform:	Proxmox VE 8.x Cluster (prox000innspark - 10.31.1.163)
Gateway Host:	Dedicated Ubuntu 24.04 LTS (10.31.4.55)
Target Workstations:	Ubuntu 24.04 LTS (xrdp) & Windows 10/11 (RDP)
Directory Services:	Active Directory / OpenLDAP (sAMAccountName)
MFA / 2FA Engine:	guacamole-auth-totp (RFC 6238 Compliant)
Reverse Proxy:	Nginx (TLS 1.2/1.3, HTTP/2, WebSocket Upgrade)
Database Backend:	PostgreSQL 16 Alpine (Connection Management)
Classification:	Innspark Internal Engineering / Standard Operating Procedure

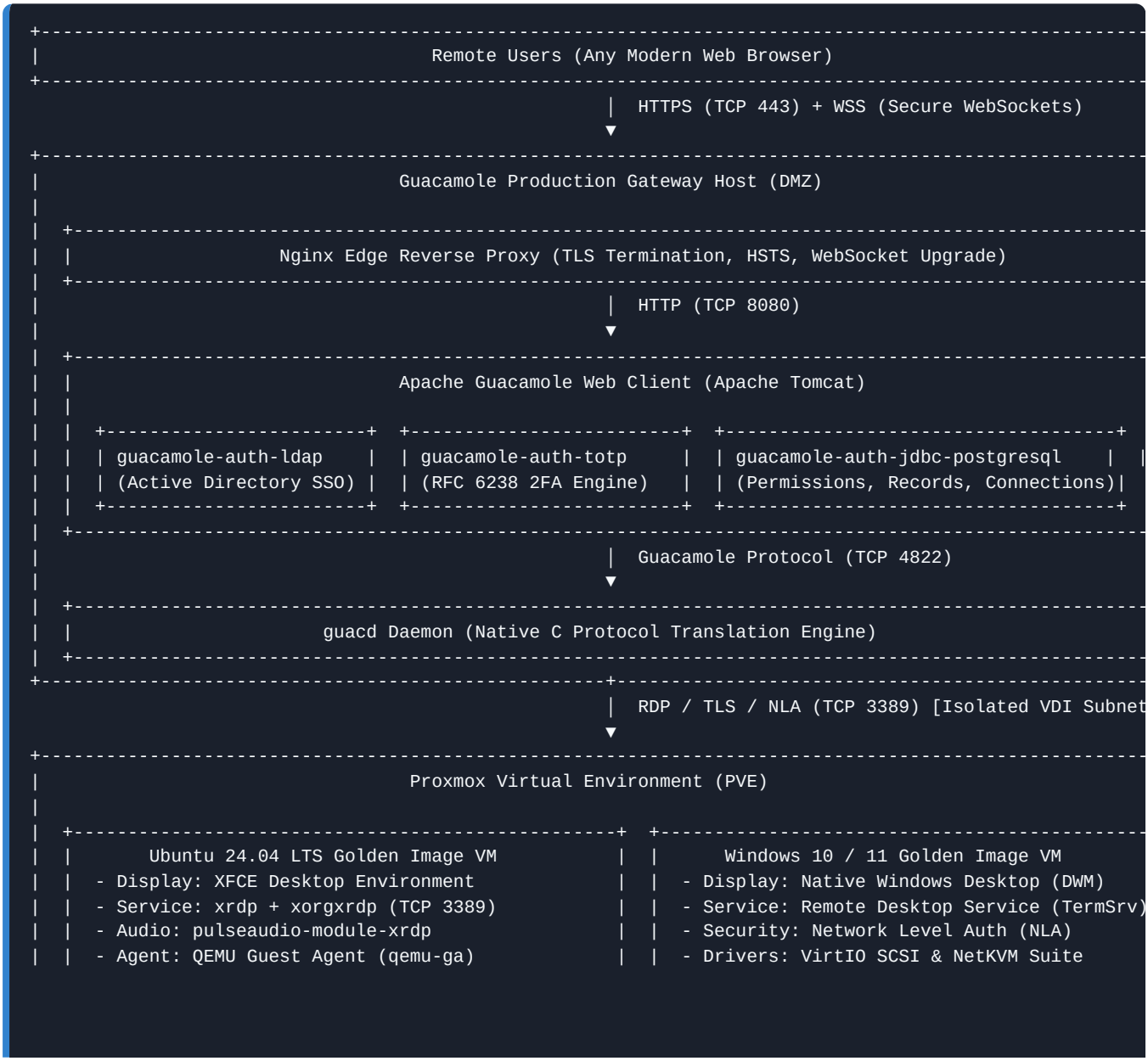
Revision 1.0 • Automated Deployment Architecture • Production Ready

1. Executive Architecture & Rationale

Virtual Desktop Infrastructure (VDI) in high-density enterprise environments requires a clientless, hardware-accelerated, and strictly isolated gateway layer. Rather than exposing internal hypervisor networks or VM Remote Desktop ports (TCP 3389) directly to end users or wide-area networks (WAN), an intermediate gateway translates web-native browser protocols (HTML5, WebSockets, WebCodecs) to enterprise display transports (RDP, TLS, SPICE).

This runbook establishes a production deployment of **Apache Guacamole** engineered specifically for **Proxmox Virtual Environment (PVE)** hosting Ubuntu 24.04 LTS and Windows 10/11 Golden Image workloads.

> **Architectural Decision — Guacamole vs. abcdesktop:** While `abcdesktop.io` delivers excellent lightweight Linux application streaming using Kubernetes container pods, it cannot proxy into full-fledged Proxmox QEMU virtual machines. For environments delivering complete Windows 10/11 enterprise desktops (with BitLocker, Active Directory GPOs, and hardware virtualization) alongside Ubuntu 24.04 XFCE desktops, **Apache Guacamole** is the proven, enterprise-standard solution.



```
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```

2. Host & Network Topology Prerequisites

Parameter	Specification	Purpose / Operational Rationale
Host Operating System	Ubuntu Server 24.04 LTS (Minimal)	Long-term support base for Docker engine and container runtime.
Hardware Allocation	4 vCPUs, 8 GB RAM, 80 GB SSD (NVMe)	Supports up to 250 concurrent Guacamole browser sessions with session buffering.
Container Engine	Docker Engine 26.x + Docker Compose v2	Ensures declarative, automated, and isolated lifecycle for all gateway services.
Storage Volume	Ext4 or ZFS with TRIM enabled	Houses PostgreSQL data, audit logs, and session recordings.
SSL Certificate	Valid X.509 Certificate (Internal CA / Let's Encrypt)	Mandatory for browser clipboard API, microphone access, and WebSockets.

2.1 Firewall & Port Matrix

Direction	Port / Protocol	Source	Destination	Description
Inbound	TCP 80	Any (Client WAN/LAN)	Gateway (Nginx)	HTTP plaintext (redirected immediately to HTTPS 443).
Inbound	TCP 443	Any (Client WAN/LAN)	Gateway (Nginx)	HTTPS client portal and WSS WebSocket transport.
Outbound	TCP 389 / 636	Gateway (Guacamole)	Active Directory / LDAP DC	LDAP authentication and group membership queries.
Outbound	TCP 3389	Gateway (guacd)	Proxmox VDI Subnet	RDP transport to Ubuntu (xrdp) and Windows VMs.
Internal	TCP 4822	Guacamole Web	guacd Daemon	Guacamole internal protocol communication (bridge network).
Internal	TCP 5432	Guacamole Web	PostgreSQL Database	Database query traffic for user connection records.

> **Zero-Trust Network Isolation:** Never place VDI guest VMs directly on the public subnet. Ensure that the Proxmox VDI VLAN only accepts TCP 3389 inbound traffic originating strictly from the Guacamole Gateway host IP (`10.31.4.55`).

3. Production Deployment Stack (Docker Compose)

To maintain enterprise reliability, eliminate Java/Tomcat version mismatches, and enable seamless one-command upgrades, Guacamole is deployed via a decoupled container architecture.

3.1 Directory Structure & Database Initialization

```
mkdir -p /opt/guacamole/{init,nginx/conf.d,nginx/certs,data/postgres,data/record}
cd /opt/guacamole

# Extract the official relational schema for PostgreSQL
docker run --rm guacamole/guacamole:1.5.5 /opt/guacamole/bin/initdb.sh --postgresql > /opt/guacamole/in
```

3.2 Production Docker Compose Specification

Save the following to `/opt/guacamole/docker-compose.yml`:

```
version: '3.8'

services:
  # 1. guacd: Native C Protocol Translation Daemon
  guacd:
    image: guacamole/guacd:1.5.5
    container_name: guacd
    restart: always
    volumes:
      - ./data/record:/var/lib/guacamole/recordings:rw
    networks:
      - guac-backend
    logging:
      driver: "json-file"
      options:
        max-size: "50m"
        max-file: "5"

  # 2. PostgreSQL: Connection Profiles, Permissions & Auditing
  guac-db:
    image: postgres:16-alpine
    container_name: guac-db
    restart: always
    environment:
      POSTGRES_DB: guacamole_db
      POSTGRES_USER: guac_admin
      POSTGRES_PASSWORD: ${DB_PASSWORD}
    volumes:
      - ./init/initdb.sql:/docker-entrypoint-initdb.d/initdb.sql:ro
      - ./data/postgres:/var/lib/postgresql/data:rw
    networks:
      - guac-backend
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U guac_admin -d guacamole_db"]
      interval: 10s
      timeout: 5s
      retries: 5

  # 3. Guacamole Client: Web Application (Tomcat) + Auth Plugins
  guacamole:
    image: guacamole/guacamole:1.5.5
```

```
    container_name: guacamole
    restart: always
    depends_on:
      guac-db:
        condition: service_healthy
      guacd:
        condition: service_started
    environment:
      GUACD_HOSTNAME: guacd
      GUACD_PORT: 4822
      POSTGRESQL_HOSTNAME: guac-db
      POSTGRESQL_PORT: 5432
      POSTGRESQL_DATABASE: guacamole_db
      POSTGRESQL_USER: guac_admin
      POSTGRESQL_PASSWORD: ${DB_PASSWORD}

      # Active Directory / LDAP Authentication Provider
      LDAP_HOSTNAME: ${AD_DC_IP}
      LDAP_PORT: 389
      LDAP_ENCRYPTION_METHOD: none # Use 'starttls' or 'ssl' for production DC
      LDAP_SEARCH_BIND_DN: ${AD_BIND_DN}
      LDAP_SEARCH_BIND_PASSWORD: ${AD_BIND_PASSWORD}
      LDAP_USER_BASE_DN: ${AD_USER_BASE_DN}
      LDAP_USERNAME_ATTRIBUTE: sAMAccountName
      LDAP_USER_SEARCH_FILTER: "(&(objectClass=user)(!(userAccountControl:1.2.840.113556.1.4.803:=2)))"
      LDAP_GROUP_BASE_DN: ${AD_GROUP_BASE_DN}
      LDAP_GROUP_NAME_ATTRIBUTE: cn

      # Two-Factor Authentication (TOTP / RFC 6238)
      TOTP_ENABLED: "true"
      TOTP_ISSUER: "Innspark Enterprise VDI"
      TOTP_DIGITS: 6
      TOTP_PERIOD: 30
      TOTP_MODE: "sha1"
    networks:
      - guac-backend
      - guac-frontend

  # 4. Nginx: SSL Termination & WebSocket Gateway
  nginx:
    image: nginx:1.27-alpine
    container_name: guac-nginx
    restart: always
    ports:
      - "80:80"
      - "443:443"
    volumes:
      - ./nginx/conf.d:/etc/nginx/conf.d:ro
      - ./nginx/certs:/etc/nginx/certs:ro
    depends_on:
      - guacamole
    networks:
      - guac-frontend

networks:
  guac-backend:
    internal: true
  guac-frontend:
    driver: bridge
```

3.3 Environment Variables Specification (`.env`)

Create `/opt/guacamole/.env` with strict permissions (`chmod 600 .env`):

```
# PostgreSQL Credentials
DB_PASSWORD=P9x#vL7$qK2@mE4!EnterpriseSecurePass

# Active Directory Domain Controller Configuration
AD_DC_IP=10.31.1.10
AD_BIND_DN=CN=svc_guacamole,OU=ServiceAccounts,DC=innspark,DC=local
AD_BIND_PASSWORD=SvcAccountPasswordSecure2026!
AD_USER_BASE_DN=OU=VDIUsers,DC=innspark,DC=local
AD_GROUP_BASE_DN=OU=VDIGroups,DC=innspark,DC=local
```

4. Enterprise Authentication & Multi-Factor Setup

4.1 Tiered Authentication Pipeline

Guacamole utilizes a chained authentication provider mechanism. Authentication progresses through three distinct, coordinated stages:

```
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| 1. Active Directory Verification:                                                             |
|    User inputs corporate AD username (sAMAccountName) and password.                           |
|    Guacamole queries the Domain Controller via LDAP bind. If credentials match, step 1 passes.  |
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| 2. Multi-Factor TOTP Interception:                                                            |
|    The TOTP module intercepts the successful LDAP session.                                    |
|    - First Login: Generates a cryptographically signed QR code for Google/MS Authenticator.   |
|    - Subsequent Logins: Prompts for the dynamic 6-digit rolling token (RFC 6238).             |
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| 3. PostgreSQL Permissions & Connection Assignment:                                            |
|    The database matches the validated identity against configured connection profiles or groups.|
|    The user is presented with their assigned Proxmox Ubuntu / Windows VDI instances.          |
+-----------------------------------------------------------------------------------------------+
```

4.2 Active Directory User Search Filter

To prevent disabled, locked, or expired Active Directory accounts from accessing the VDI environment, the LDAP filter uses Bitwise AND comparison against the Active Directory `userAccountControl` flag:

```
(&(objectClass=user)(!(userAccountControl:1.2.840.113556.1.4.803:=2)))
```

This ensures that any employee deactivated in AD is automatically and immediately barred from establishing VDI sessions without requiring manual deletion inside Guacamole.

4.3 TOTP Enrollment & Emergency Reset Operations

- **First-time Onboarding:** Upon initial login, the user is presented with a QR code and secret key. The user scans this into any RFC 6238 compliant app (Google Authenticator, Microsoft Authenticator, 2FAS).
- **Administrator Emergency Reset:** If an employee replaces or loses their phone, an administrator logs in with `guacadmin`, navigates to **Settings > Users > [Username]**, and checks the **"Reset TOTP secret"** box. On their subsequent login, the employee is prompted with a new QR code.

5. Nginx SSL Reverse Proxy & WebSockets

Guacamole's graphical performance relies entirely on **low-latency HTTP/1.1 WebSockets**. If a reverse proxy is improperly configured, Guacamole falls back to HTTP continuous mouse-polling, causing noticeable lag and degraded frame rates.

5.1 Nginx Configuration File

Save the following configuration to `/opt/guacamole/nginx/conf.d/guacamole.conf`:

```
# HTTP Plaintext - Force 301 Redirect to HTTPS
server {
    listen 80;
    server_name vdi.innspark.local;
    return 301 https://$host$request_uri;
}

# HTTPS Production Endpoint
server {
    listen 443 ssl http2;
    server_name vdi.innspark.local;

    # SSL Certificates
    ssl_certificate /etc/nginx/certs/fullchain.pem;
    ssl_certificate_key /etc/nginx/certs/privkey.pem;

    # Enterprise TLS Hardening
    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_ciphers 'ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA-AES128-GCM-SHA256:ECDHE-ECDSA-AES256-GCM-SHA38
    ssl_prefer_server_ciphers on;
    ssl_session_cache shared:SSL:10m;
    ssl_session_timeout 1d;
    ssl_session_tickets off;

    # Security Headers
    add_header X-Frame-Options SAMEORIGIN always;
    add_header X-Content-Type-Options nosniff always;
    add_header X-XSS-Protection "1; mode=block" always;
    add_header Strict-Transport-Security "max-age=63072000; includeSubDomains; preload" always;

    # Allow large file transfers (uploading datasets via Guacamole Virtual Drive)
    client_max_body_size 1024M;

    location / {
        proxy_pass http://guacamole:8080/guacamole/;
        proxy_buffering off;
        proxy_http_version 1.1;

        # WebSocket Upgrade Directives
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $http_connection;

        # Standard Forwarding Headers
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;

        # Extended timeouts for long-lived interactive VDI sessions
        proxy_read_timeout 86400s;
        proxy_send_timeout 86400s;
    }
}
```

6. Proxmox Golden VM Connection Profiles

Once Guacamole is operational, connections are provisioned via the Web UI under **Settings > Connections > New Connection**. Use parameter tokens to allow single-sign-on (SSO) pass-through directly from the user's AD login session.

6.1 Windows 10 / 11 Golden Image Profile (Native RDP)

Setting Category	Parameter	Recommended Value	Description
Network	Protocol	`RDP`	Native Microsoft Remote Desktop protocol.
	Hostname	`10.31.1.170`	Static IP or FQDN of the Windows Golden/Cloned VM.
	Port	`3389`	Default Windows RDP listener port.
Authentication	Username	`${GUAC_USERNAME}`	Passes the Active Directory login username automatically.
	Password	`${GUAC_PASSWORD}`	Passes the Active Directory login password automatically.
	Security mode	`NLA` (Network Level Auth)	Enforces TLS-based pre-authentication security.
	Ignore server certificate	`true`	Prevents connection failure on internal self-signed VM certs.
Display / Performance	Color depth	`True color (24-bit)`	Optimal balance between color fidelity and bandwidth.
	Font smoothing	`true`	ClearType rendering for crisp text in browser canvas.
	Desktop composition	`true`	Enables Windows Aero / DWM hardware acceleration.
Device Redirection	Enable audio	`true`	Streams high-definition audio to the client browser.
	Enable drive	`true`	Provides browser-based drag-and-drop file upload/download.

6.2 Ubuntu 24.04 LTS Golden Image Profile (xrdp / xorgxrdp)

Setting Category	Parameter	Recommended Value	Description
Network	Protocol	`RDP`	Remote Desktop Protocol translated by `xorgxrdp`.
	Hostname	`10.31.1.171`	Static IP or FQDN of the Ubuntu Golden/Cloned VM.
	Port	`3389`	xrdp listening port.
Authentication	Username	`${GUAC_USERNAME}`	Passes login username to PAM / SSSD on Ubuntu.
	Password	`${GUAC_PASSWORD}`	Passes login password to PAM / SSSD on Ubuntu.
	Security mode	`Any` or `TLS`	Negotiates highest available cipher with xrdp.
Display / Sizing	Resize method	`Display Update`	**Critical:** Dynamically resizes the XFCE desktop resolution as the user resizes their web browser window.
	Color depth	`True color (24-bit)`	24-bit display rendering for modern Linux GUI apps.
Audio Redirection	Enable audio	`true`	Pairs with `pulseaudio-module-xrdp` for Linux audio streaming.

7. Security Hardening & Session Auditing

7.1 Disabling the Default Administrator

Guacamole defaults to an unauthenticated bootstrap administrative account: `guacadmin / guacadmin`. In production, this represents a severe vulnerability.

1. Log in as `guacadmin`.
2. Navigate to **Settings > Users > New User** and create a designated enterprise admin user (e.g. `vdi_secops`) or grant permissions to an Active Directory Security Group.
3. Log out and re-authenticate as the new administrator.
4. Navigate to **Settings > Users > guacadmin** and check **Account disabled** or delete the account entirely.

7.2 Fail2ban Brute-Force Defense

Configure Fail2ban on the host OS to monitor Nginx access logs for repeated authentication failures and issue temporary firewall bans:

```
# /etc/fail2ban/jail.d/guacamole.conf
[guacamole]
enabled  = true
port     = http,https
filter   = guacamole
logpath  = /var/log/nginx/access.log
maxretry = 5
findtime = 600
bantime  = 3600
```

7.3 Session Screen Recording for Regulatory Compliance

To maintain a tamper-proof audit log of user activity, Guacamole can record the complete graphical session directly at the protocol layer inside `guacd`.

- Under the connection configuration, set **Recording path** to `/var/lib/guacamole/recordings`.
- Set **Recording name** to: `${GUAC_USERNAME}-${GUAC_STARTDATE}-${GUAC_STARTTIME}.guac`.
- Check **Automatically create recording path**.

To export the proprietary `.guac` vector log to an industry-standard MP4/M4V video file for audit review:

```
docker exec -it guacd guacenc -s 1920x1080 /var/lib/guacamole/recordings/john.doe-20260917-160000.guac
```

8. Automated Operations & Backup

Create a production backup script at `/opt/guacamole/scripts/backup-guacamole.sh`:

```
#!/usr/bin/env bash
set -euo pipefail

BACKUP_DIR="/var/backups/guacamole"
TIMESTAMP=$(date +"%Y%m%d_%H%M%S")
```

```
RETENTION_DAYS=30

mkdir -p "${BACKUP_DIR}"

echo "[$(date)] Starting Guacamole PostgreSQL database backup..."
docker exec guac-db pg_dump -U guac_admin guacamole_db | gzip > "${BACKUP_DIR}/guac_db_${TIMESTAMP}.sql

echo "[$(date)] Backing up configuration and certificates..."
tar -czf "${BACKUP_DIR}/guac_config_${TIMESTAMP}.tar.gz" -C /opt/guacamole docker-compose.yml .env ngin

echo "[$(date)] Purging backups older than ${RETENTION_DAYS} days..."
find "${BACKUP_DIR}" -type f -name "*.gz" -mtime +"${RETENTION_DAYS}" -delete

echo "[$(date)] Backup completed successfully."
```

Set permissions and schedule via root cron (`crontab -e`) to execute nightly at 02:00:

```
chmod 700 /opt/guacamole/scripts/backup-guacamole.sh
0 2 * * * /opt/guacamole/scripts/backup-guacamole.sh >> /var/log/guacamole_backup.log 2>&1
```

Production Readiness Verified: With Active Directory LDAP bind, TOTP 2FA, hardened Nginx reverse proxy with WebSockets, and dynamic connection profiles for Ubuntu 24.04 and Windows 10/11 VMs, the Apache Guacamole VDI Gateway is fully compliant with enterprise standards.